

Introduction to Modeling for Formal Verification

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Today's goals


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Module 1 ⤴

Introduction to Model Checking

Motivation of Verification

- ✔ Course Updates and Accessibility Support
Reading • 1 min
- ✔ Non-Credit Students: Welcome and Where to Find Help
Reading • 10 min
- ✔ Motivating Examples
Video • 7 min
- ✔ High Profile Previous Bugs and Their Impacts
Video • 10 min
- ✔ Principles of Model Checking - Chapter 1, Introduction - Part 1
Reading • 15 min
- ✔ Hardware and Software Bugs - Wikipedia
Reading • 20 min
- ✔ Motivation of Verification Quiz
Graded Assignment • Grade: 100%

Overview of Verification

- ✔ Overview of Functional Verification
Video • 5 min
- ✔ Techniques for Functional Verification
Video • 10 min
- ✔ Functional Verification
Reading • 30 min
- ✔ Principles of Model Checking - Chapter 1, Introduction - Part 2
Reading • 15 min
- ✔ Overview of Verification Quiz
Graded Assignment • Grade: 100%

Formal Verification and Model Checking

- ✔ Overview of Formal Verification and Model Checking
Video • 7 min
- ✔ Elements of Model Checking
Video • 6 min
- ✔ Challenge to Model Checking: State Space Explosion
Video • 5 min
- ✔ Principles of Model Checking - Section 1.1, 1.2
Reading • 30 min
- Formal Verification and Model Checking - Wikipedia
Reading • 30 min
- Formal Verification and Model Checking Quiz
Graded Assignment • 10 min

Module 2 ⤴

Modeling

Module 3 ⤴

Modeling System Composition

Formal Verification and Model Checking - Wikipedia



Formal Verification - Wikipedia Page

The Wikipedia page offers additional information about formal verification. It also contains links to other wiki pages for information about different techniques for formal verification.

Read about formal verification at the following link: [Formal Verification - Wikipedia Page](#) (external link; opens in new tab).

Model Checking - Wikipedia Page

This Wikipedia page offers additional information about model checking, including some significant developments in history, and other well-known techniques for scalable model checking. It also contains a list of well-known model checking tools.

Read about model checking at the following link: [Model Checking - Wikipedia Page](#)  (external link; opens in a new tab).

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